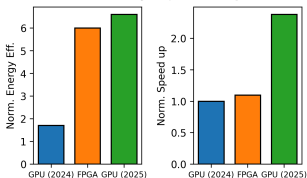
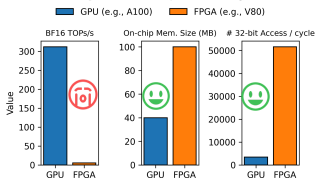


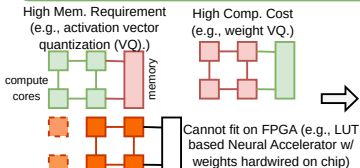
Motivation 1: Growing performance of GPU w/ only algorithm updates



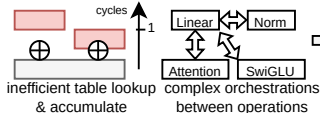
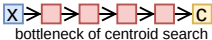
Motivation 2: FPGA vs. GPU - Low Compute but High Memory



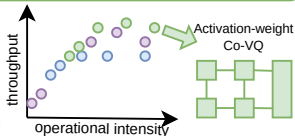
Promising Goal: Improve LLM inference efficiency on FPGA with memory based operations to exploit the abundant and distributed on-chip memory resources.



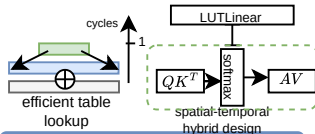
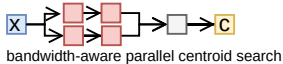
Challenge 1: Inefficient/Impossible when scale to language models



Challenge 2: Additional Complexity of LLMs w/ memory-based operations



Contri. 1: Performance Model & New Efficient Algorithm (Sec. 3)



Contri. 2: Specialized Arch. for LLM w/ memory-based operations (Sec. 4)